

Investigating Robust Overfitting in Adversarial Training

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Adversarial Example: Panda to Gibbon (Goodfellow et al., 3)

1. We see a panda, the model sees a panda as well.
2. We add noises to the image, invisible to human eyes.
3. We still see a panda, but the model sees something else: a gibbon.



x

“panda”

57.7% confidence

+ .007 ×



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

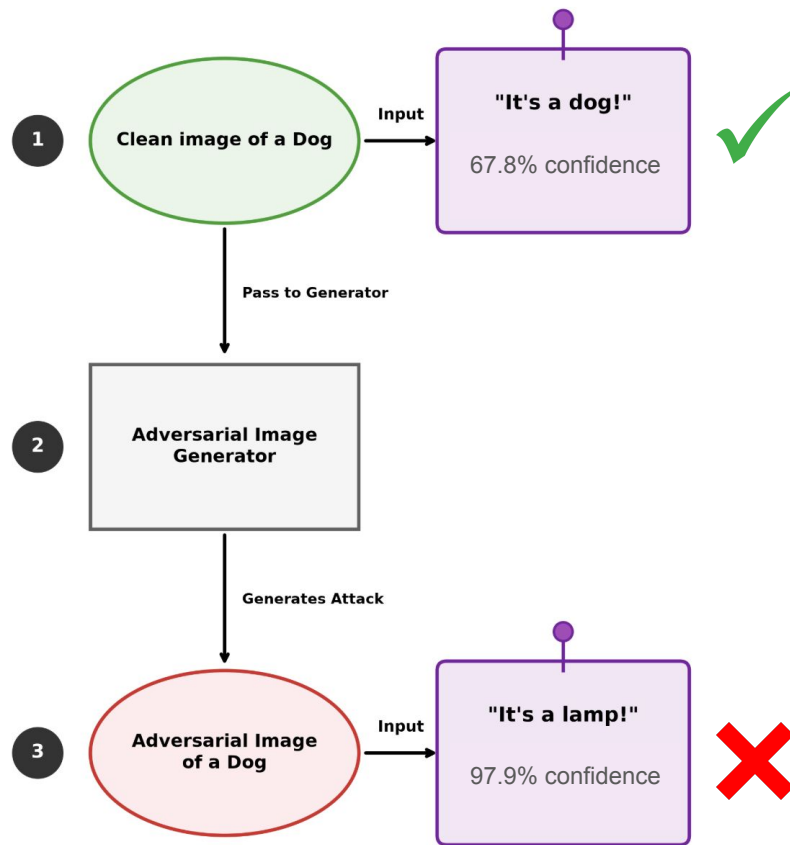
99.3 % confidence

Definition (Goodfellow et al., 1)

Adversarial Attack: An imperceptible input perturbation that tricks neural networks into high-confidence misclassifications.

Adversarial Image Generator: A method that computes the smallest input change needed to fool the model (more details on next slide).

Adversarial Image: Normal-looking picture that has been intentionally changed to trick a model into misclassifying what it sees confidently.



Generate Adversarial Images with FGSM

$$\eta = \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)), \quad \tilde{x} = x + \eta$$

where $\mathbf{x}$: clean image input,

ϵ : size of allowed change,

η : the perturbation,

$\tilde{\mathbf{x}}$: adversarial image output.

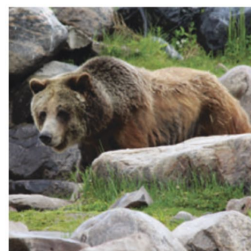
How It Works:

1. Compute the gradient of model's loss with respect to input image
2. Take the sign of that gradient, just +1 or -1 per pixel, direction only
3. Scale by ϵ to keep the change small and invisible
4. Add the result to the original image to get the adversarial image $\tilde{x}$, our output

Other Adversarial Examples

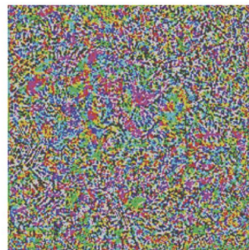
Example 2: Bear to Truck

illustration of a typical adversarial example
ANN classification: bear



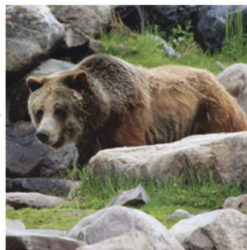
64x

+



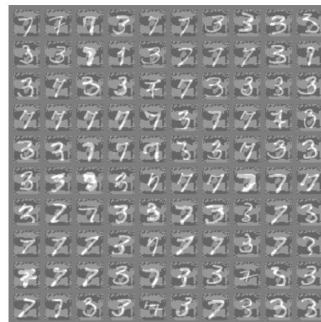
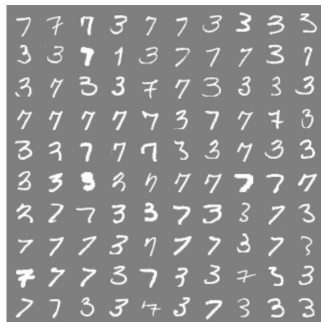
→

ANN classification: truck



(Veerabadran et al., 2)

Example 3: MNIST Logistic Regression



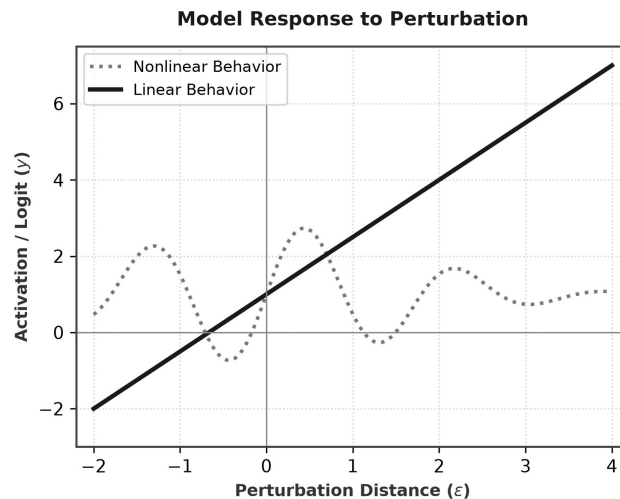
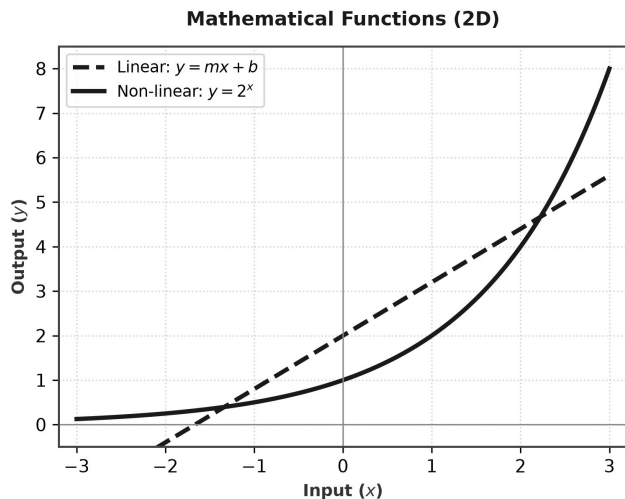
(Goodfellow et al., 4)

Why Does This Happen? (Goodfellow et al., 1)

First assumption: Caused by model behaving nonlinearly (unpredictable).

Actual root cause: Model behaving linearly (predictable), especially in high-dimensional spaces.

But how? Because predictable also means it is exploitable.

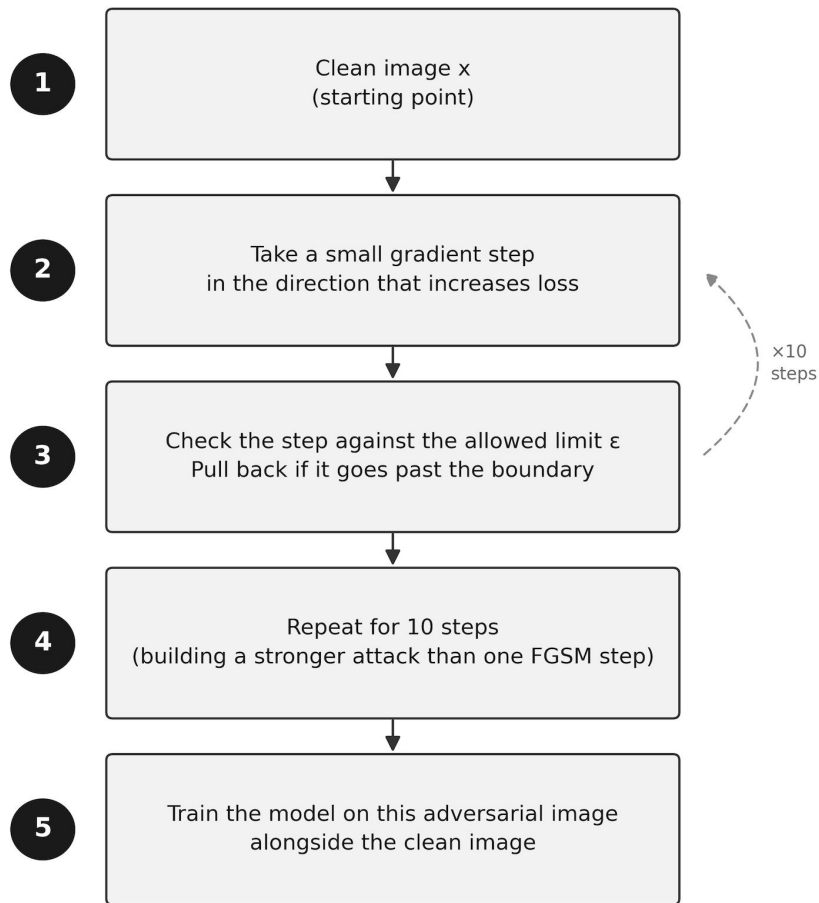


How to Defend? (Rice et al., 3-4)

Adversarial Training: Train on both clean and adversarial images.

PGD (Projected Gradient Descent) vs. FGSM (Fast Gradient Sign Method): FGSM takes one step. PGD takes several small steps, checking the boundary each time. Rice et al. use PGD rather than FGSM.

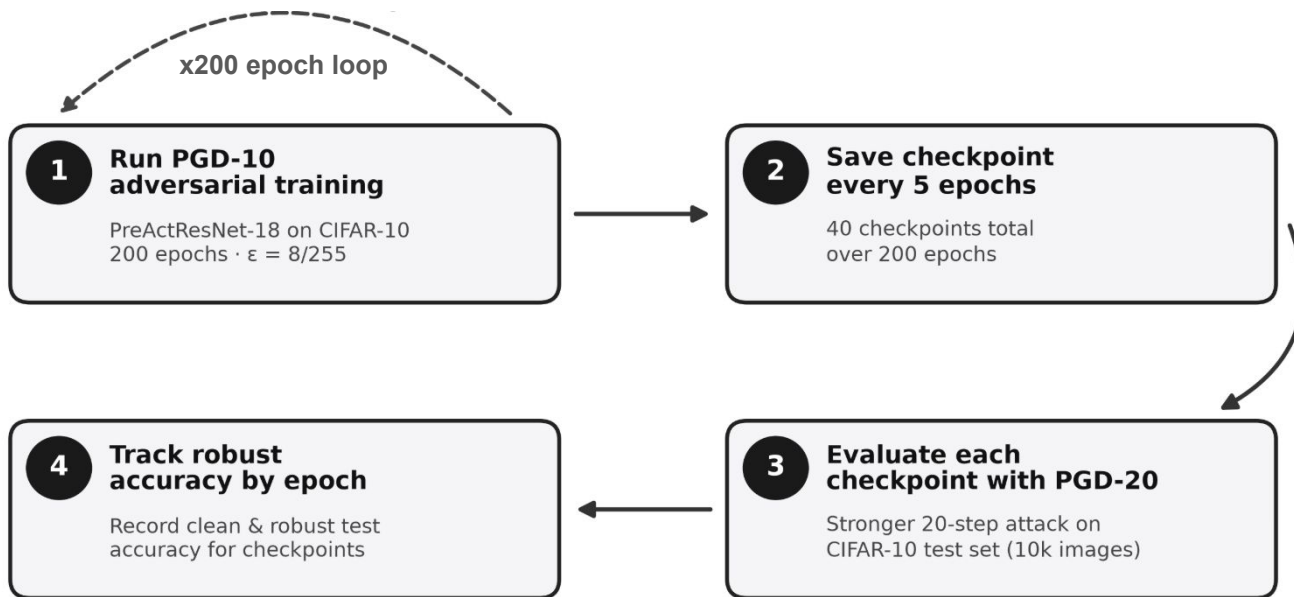
Step-by-step on how PGD adversarial training works is shown on right side.



Adversarial Training (Rice et al., 4)

Neural Network Architecture: PreActResNet-18

Dataset: CIFAR-10 · 50,000 images for training · 10,000 for testing



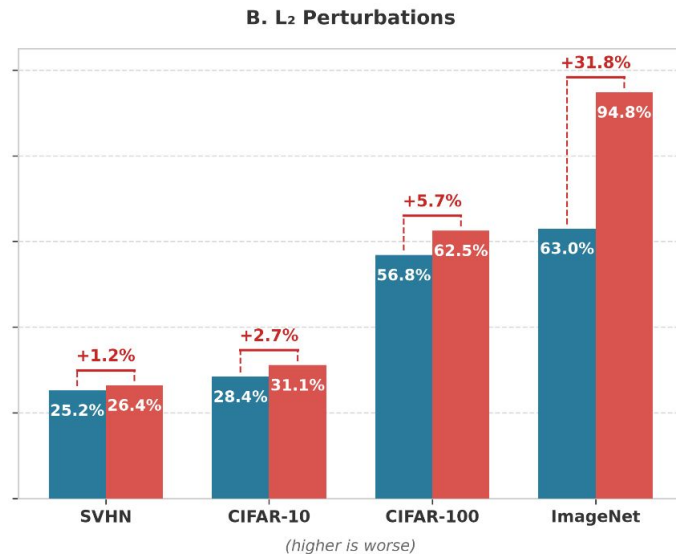
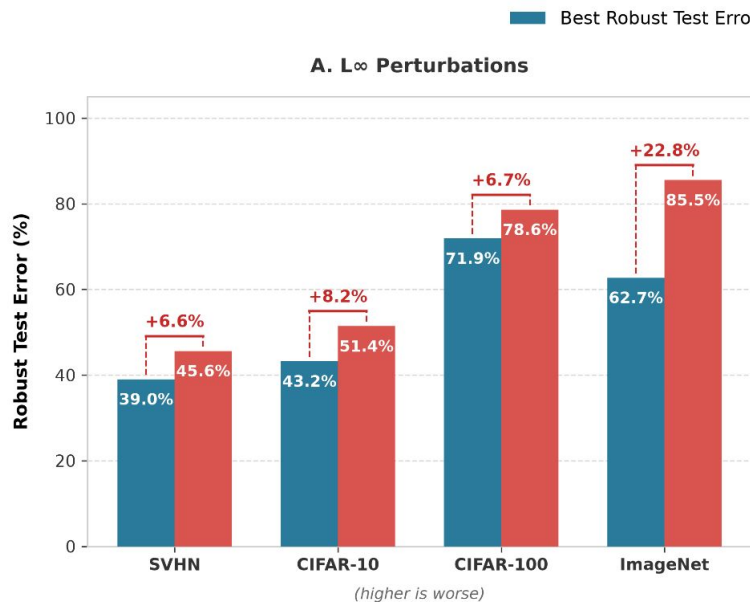
Training Too Long Can Hurt Robustness (Rice et al., 4-5)

Robust overfitting:

1. Robustness peaks early
2. Continued training erodes those gains

Robust test error: % of attacked test images misclassified.

Training longer does not mean more robust.

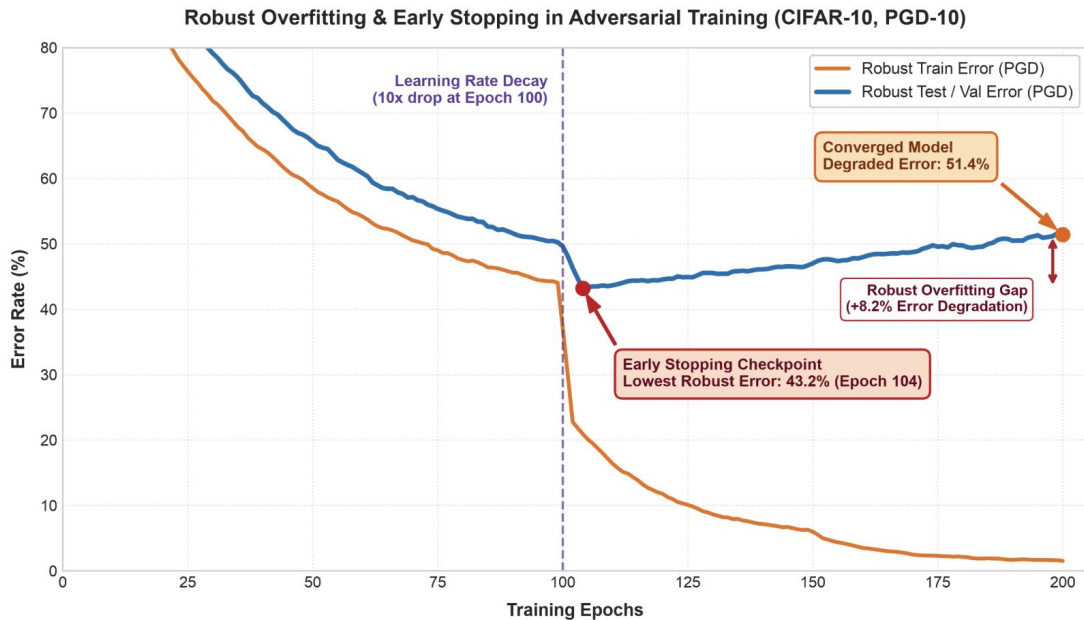


Solution: Early Stopping (Rice et al., 1, 6)

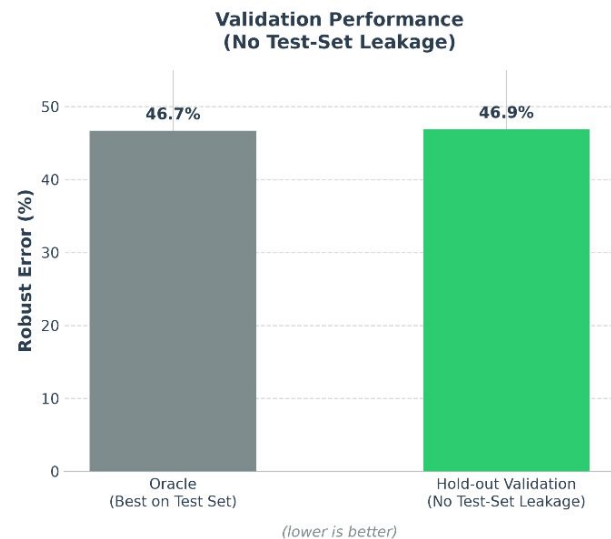
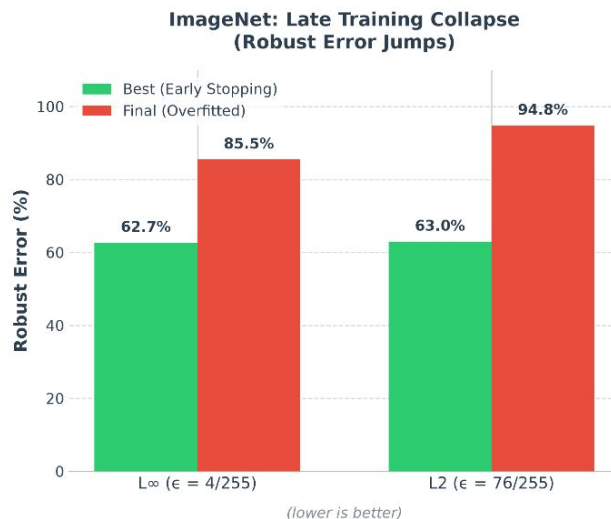
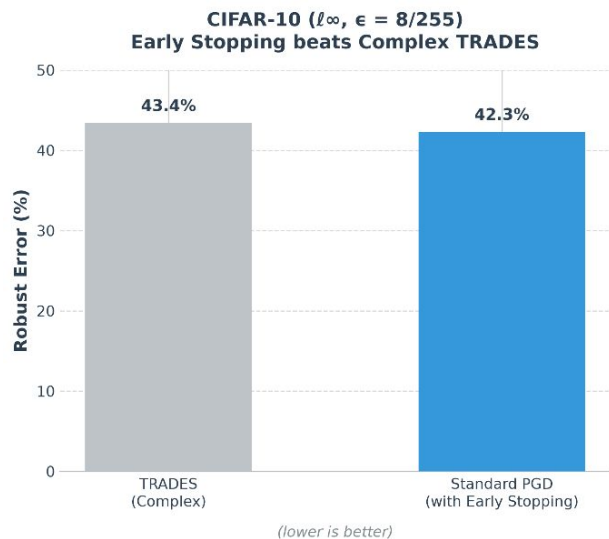
Core Mechanism: Save the best checkpoint evaluated on a held-out adversarial validation set rather than training until convergence.

Preventing Degradation: Halt training shortly after initial learning rate decay to avoid severe robust test set overfitting.

Proven Effectiveness: Simple early stopping matches or outperforms complex explicit regularization techniques across architectures.



Effectiveness of Early Stopping (Rice et al., 4)



Our Research: Motivation and Scope

Our Research Question:

Why does robust overfitting happen?
This was not stated in Rice et al.
paper.

Hypothesis:

Models memorize specific
perturbation patterns instead of
learning a boundary that generalizes
to new attacks.

This Summer Research:

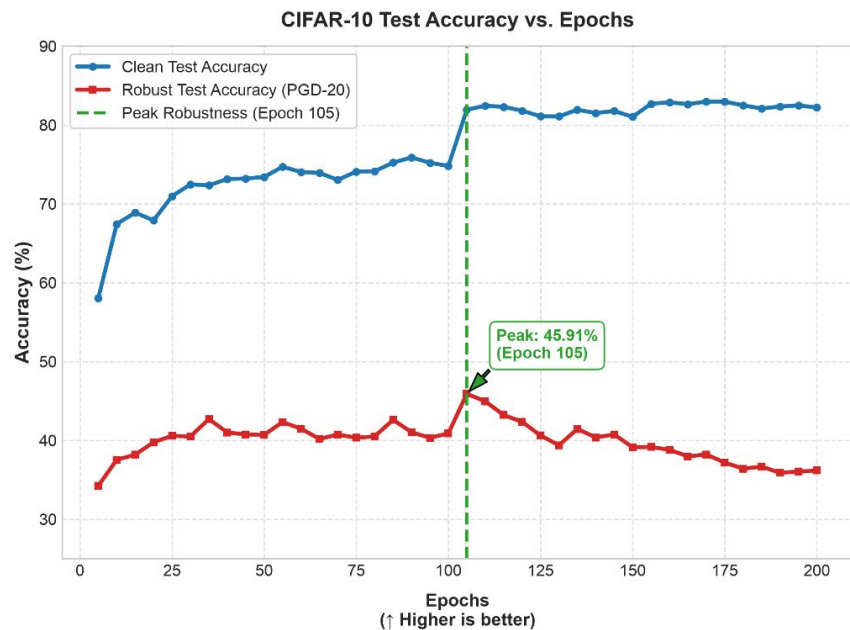
Reproduce Rice et al.'s findings on
CIFAR-10 and precisely locate where
the overfitting gap emerges.

Why This Matters:

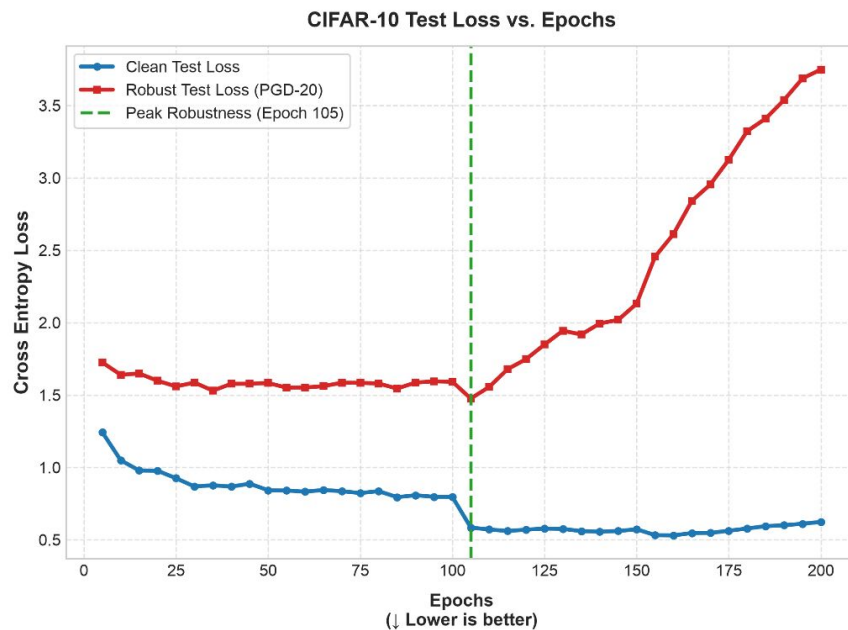
Knowing the cause could replace
Early Stopping with a better fix,
potentially making models much
stronger against adversarial attacks.

Test Results Under PGD-20 on CIFAR-10

Robust accuracy peaks at Epoch 105, then degrades with more training.



Robust loss stays stable until Epoch 105, then more than doubles by Epoch 200.



Training and Test Results Under Different Seeds

[Text goes here]

All code for reproducing the experiments as well as pretrained model weights and training logs can be found at GitHub repository: github.com/ltsKaiwenDu/Robust-Overfitting.

Key Takeaways

1. Adversarial examples, which can be generated with FGSM, are caused by linearity (predictable so exploitable), not nonlinearity (unpredictable), to reliably trick confident models.
2. Adversarial training defends against adversarial attacks by training on both clean and perturbed examples.
3. Robust overfitting causes adversarially trained models to get worse on new data the longer they train.
4. Early stopping is the best and simplest way to resist robust overfitting and this method matches or beats more complex methods.
5. [Reserved for our research findings]
6. [Reserved for our research findings]

References

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and Harnessing Adversarial Examples." *International Conference on Learning Representations (ICLR)*, 2015.

Rice, Leslie, Eric Wong, and J. Zico Kolter. "Overfitting in Adversarially Robust Deep Learning." Proceedings of the 37th International Conference on Machine Learning, *PMLR*, vol. 119, 2020.

Veerabadran, Vijay, et al. "Subtle Adversarial Image Manipulations Influence Both Human and Machine Perception." *Nature Communications*, vol. 14, no. 1, 2023, article 4933. Springer Nature, 15 Aug. 2023, <https://doi.org/10.1038/s41467-023-40499-0>

Du, Kaiwen, and Nicholas Q. Tran. Investigating Robust Overfitting in Adversarial Training. *GitHub*, 2026, github.com/ItsKaiwenDu/Robust-Overfitting.



Thank you!

Any Questions?

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Science